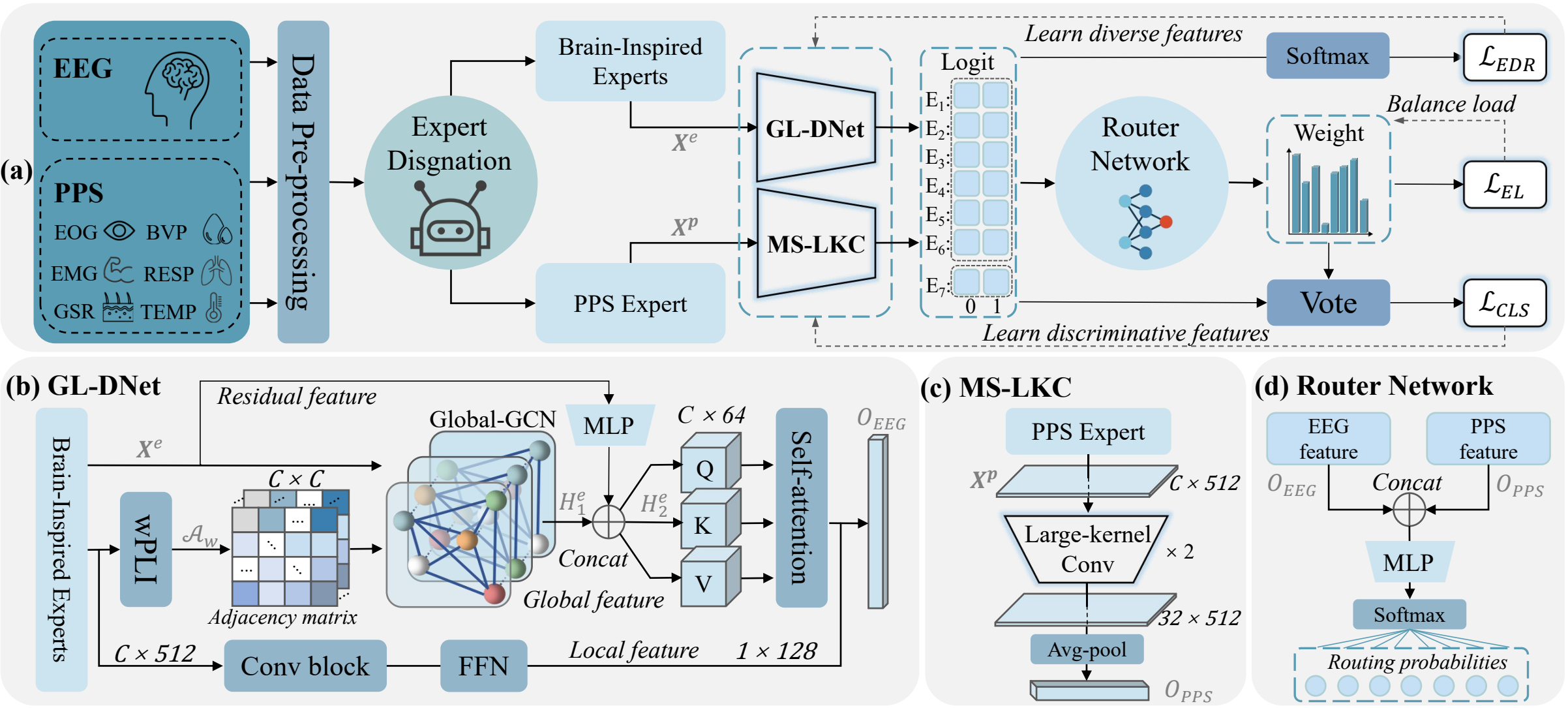
(a)
EEG
PPS
EOG
BVP
EMG
RESP
GSR
TEMP
Data Pre-processing
Expert Disgnation
Brain-Inspired Experts
PPS Expert
X^e
X^p
GL-DNet
MS-LKC
Logit
E_1
E_2
E_3
E_4
E_5
E_6
E_7
0 1
Learn diverse features
Softmax
$\mathcal{L}_{EDR}$
Balance load
Router Network
Weight
$\mathcal{L}_{EL}$
Vote
$\mathcal{L}_{CLS}$
Learn discriminative features
(b) GL-DNet
Brain-Inspired Experts
X^e
Residual feature
wPLI
$\mathcal{A}_w$
$C \times C$
Adjacency matrix
Global-GCN
MLP
H_1^e
H_2^e
Concat
Global feature
$C \times 64$
Q
K
V
Self-attention
O_{EEG}
$C \times 512$
Conv block
FFN
Local feature
1×128
(c) MS-LKC
PPS Expert
X^p
$C \times 512$
Large-kernel Conv
$\times 2$
32×512
Avg-pool
O_{PPS}
(d) Router Network
EEG feature
PPS feature
O_{EEG}
Concat
O_{PPS}
MLP
Softmax
Routing probabilities